STAT 112: Introduction to Statistics and Probability II

Lecture 1: Set Theory and Counting Processes

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Purpose and Learning Outcomes

The purpose of this lecture is to equip students with the knowledge and skills required to work with **set theory** and **counting processes**: their concepts, properties, and real-world applications.

By the end of this lecture, you should be able to:

1. Define and explain basic concepts in set theory: sample space, events, and outcomes of a random experiment.
2. Define and apply set operations (union, intersection, complement, difference, symmetric difference).
3. Use Venn diagrams to analyse two-set and three-set problems.
4. Apply the Inclusion–Exclusion Principle to count elements in overlapping sets.
5. Compute the number of ordered arrangements using **permutations** (with and without restrictions).
6. Compute the number of unordered selections using **combinations** (with and without restrictions).
7. Solve applied counting problems drawn from Ghanaian and international contexts.

Lecture Outline

1. 1. Introduction to Random Experiments
2. 2. Sets: Definitions and Notation
3. 3. Set Operations and Their Properties
4. 4. Venn Diagrams and Inclusion–Exclusion
5. 5. Counting Principles
6. 6. Permutations
7. 7. Combinations
8. 8. Summary of Key Formulae

What is a Random Experiment?

In everyday life, many activities have outcomes we *cannot predict in advance*: whether a coin shows heads, which team wins a football match, or how many customers visit a bank on a given day. Such activities are formalised through the notion of a **random experiment**.

Definition: Random Experiment

A **random experiment** is any process or activity satisfying **all three** of the following:

(i) The experiment can, in principle, be *repeated under identical conditions*.

(ii) All possible outcomes are *known in advance* (before the experiment is performed).

(iii) The exact outcome of any particular trial *cannot* be predicted with certainty.

The third condition is what makes the experiment **random**: there is genuine uncertainty. Computing $2 + 2$ always gives 4: that is *deterministic*, not random. By contrast, asking "which face appears when a die is rolled?" satisfies all three conditions.

Examples of Random Experiments

Classic Probability Settings:

- Rolling a fair six-sided die and observing which face appears.
- Tossing a coin and noting Head (H) or Tail (T).
- Drawing a card at random from a well-shuffled standard deck of 52 cards.

Continuous / Measurement Settings:

- Recording the daily high temperature in Accra on a randomly selected day.
- Measuring the waiting time (in minutes) at a Mobile Money kiosk in Madina Market.

Count / Discrete Settings:

- Observing the daily number of loan applications received at a bank in Tema.

Survey Settings:

- Selecting a student at random from a UGRC 110 class of 800 and recording their CGPA.
- Noting whether a randomly selected voter in Kumasi voted in the last election.

Key Insight

Random does **not** mean “no pattern”: it means the *individual* outcome is uncertain, even if long-run behaviour follows a law.

Outcomes, Sample Spaces, and Events

Definition: Outcome / Sample Point

An **outcome** (also called a **sample point**) is a single, indivisible possible result of a random experiment.

Definition: Sample Space

The **sample space** S is the set of *all* possible outcomes. It is also called the **universal set** of the experiment.

Definition: Event

An **event** is any subset $A \subseteq S$ of the sample space.

Types of events:

- **Simple (elementary)** event: contains exactly *one* outcome.
- **Compound** event: contains two or more outcomes.
- **Impossible event** $\emptyset$: never occurs: $P(\emptyset) = 0$.
- **Certain event** S: always occurs: $P(S) = 1$.

Example 1.1: Single Die Roll I

Roll a fair six-sided die once.

Sample Space and Events

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$.

Event	**Set**	**Type**
Even number	$A = \{2, 4, 6\}$	Compound
Greater than 4	$B = \{5, 6\}$	Compound
The number 3	$C = \{3\}$	Simple
Greater than 6	$D = \emptyset$	Impossible

Example 1.1: Single Die Roll II

Classical Probability Formula

When all $n(S)$ outcomes are **equally likely**:

$$P(A) = \frac{n(A)}{n(S)}$$

So $P(A) = \frac{3}{6} = \frac{1}{2}, \quad P(B) = \frac{2}{6} = \frac{1}{3}, \quad P(C) = \frac{1}{6}, \quad P(D) = 0.$

Example 1.2 & 1.3: Coin Tosses

Example 1.2: Two fair coin tosses (H = Head, T = Tail)

$$S = \{HH,\ HT,\ TH,\ TT\}, \qquad n(S) = 4.$$

- A = "at least one Head" $= \{HH, HT, TH\}$, $\quad P(A) = \frac{3}{4}$.
- B = "both outcomes the same" $= \{HH, TT\}$, $\quad P(B) = \frac{1}{2}$.
- C = "first toss is a Head" $= \{HH, HT\}$, $\quad P(C) = \frac{1}{2}$.
- D = "no Heads" $= \{TT\}$, $\quad P(D) = \frac{1}{4}$.

Example 1.3: Three fair coin tosses

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad n(S) = 8.$$

- A = "exactly two *consecutive* Heads" $= \{HHT, THH\}$, $P(A) = \frac{2}{8} = \frac{1}{4}$.
- B = "first toss is a Head" $= \{HHH, HHT, HTH, HTT\}$, $P(B) = \frac{4}{8} = \frac{1}{2}$.
- C = "at least one Head and at least one Tail", $n(C) = 6$, $P(C) = \frac{6}{8} = \frac{3}{4}$.

Example 1.4: Tossing Two Dice

Two fair dice are rolled, giving $n(S) = 36$ equally likely ordered pairs (i, j).

Event	Outcomes	Prob.
A: sum equals 7	$\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$	$\frac{6}{36} = \frac{1}{6}$
B: doubles	$\{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$	$\frac{6}{36} = \frac{1}{6}$
C: sum exceeds 10	$\{(5,6),(6,5),(6,6)\}$	$\frac{3}{36} = \frac{1}{12}$
D: sum equals 2	$\{(1,1)\}$	$\frac{1}{36}$

Tip: Complementary Probability

It is often easier to compute $P(A^c)$ and use $P(A) = 1 - P(A^c)$.
e.g. $P(\text{sum} \neq 7) = 1 - \frac{1}{6} = \frac{5}{6}$.

Discrete and Continuous Sample Spaces

Definition

- A sample space is **discrete** if it contains a *finite* or *countably infinite* number of outcomes, e.g. $S = \{H, T\}$ or $S = \{0, 1, 2, \ldots\}$.
- A sample space is **continuous** if it contains an *interval* (possibly infinite) of real numbers, e.g. $S = \{x : x \geq 0\}$.

Example 1.5: Classify each sample space:

a. $S = \{0, 1, 2, 3, \ldots\}$ (calls to MTN call centre/hour): *Discrete, countably infinite.*
b. $S = \{x : 0 \leq x \leq 100\}$ (body weight in kg): *Continuous.*
c. $S = \{0, 1, 2, 3, 4, 5\}$ (defective items in a batch of 5): *Discrete, finite.*
d. $S = \{(x, y) : x \geq 0, y \geq 0\}$ (random point in first quadrant): *Continuous.*
e. $S = \{k : k \text{ is a positive odd integer}\} \cup \{4, 6\}$: *Discrete (countably infinite).*
f. $S = \{x : 0 \leq x < \infty\}$ (lifetime in years of a light bulb): *Continuous.*

Example 1.6: Classical Probability with Three-Coin Toss

Using $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, $n(S) = 8$:

Applying $P(A) = \dfrac{n(A)}{n(S)}$

- $P(\text{exactly 2 Heads}) = \dfrac{3}{8}$ (outcomes: HHT, HTH, THH).
- $P(\text{at least 1 Tail}) = \dfrac{7}{8}$ (all outcomes except HHH).
- $P(\text{all Heads}) = \dfrac{1}{8}$ (only HHH).

Complementary Counting Check

$P(\text{at least 1 Tail}) = 1 - P(\text{no Tails}) = 1 - P(HHH) = 1 - \frac{1}{8} = \frac{7}{8}$. ✓
Always verify: $P(A) + P(A^c) = 1$.

Case Study 1: Quality Control at a Ghanaian Manufacturing Firm I

Setup

A small factory in **Tema** produces batches of 10 mobile phone chargers. A quality inspector selects **2 chargers at random** (without replacement) from each batch. A particular batch contains **3 defective** and **7 non-defective** chargers.

Solution:

- $n(S) = \binom{10}{2} = \dfrac{10!}{2!\,8!} = 45$ equally likely ways to choose 2 chargers from 10.
- Let $D =$ "at least one defective charger is selected".
- It is easier to count $D^c =$ "both chargers are non-defective":

$$n(D^c) = \binom{7}{2} = 21.$$

Case Study 1: Quality Control at a Ghanaian Manufacturing Firm II

- By the complementary rule:

$$P(D) = 1 - P(D^c) = 1 - \frac{21}{45} = \frac{24}{45} = \frac{8}{15} \approx 0.533.$$

Interpretation

The inspector finds at least one defective charger in approximately **53.3%** of batches. This type of calculation is fundamental to **statistical quality control**.

Section 1 Exercises

1. Explain what is meant by a *random experiment* and give three examples from daily life in Ghana (different from those in this section).
2. A bag contains 3 red balls and 2 blue balls. One ball is drawn at random.
   (a) List the sample space. (b) Let $A =$ "a red ball is drawn". List A and find $P(A)$.
3. An experiment consists of rolling two fair dice simultaneously.
   (a) Find event A: sum equals 7. (b) Find B: both faces the same.
   (c) Find C: sum exceeds 10. (d) Verify $P(A) + P(A^c) = 1$.
4. A student is selected at random from a class of 30 (18 females, 12 males). Find the probability that the selected student is (a) female, (b) male.
5. Classify each sample space as discrete or continuous and justify briefly:
   (a) Number of patients visiting a hospital in a day.
   (b) Time (hours) a student studies per day.
   (c) Score on a 100-mark examination.
   (d) Exact rainfall (mm) in Accra in a month.
6. (*Case Study extension*) A factory batch contains 4 defective and 6 good items. An inspector picks 3 items at random. Find the probability that: (a) all 3 are good; (b) exactly 1 is defective; (c) at least 1 is defective.

What is a Set?

A **set** is a *well-defined* collection of distinct objects called **elements** or **members**. Sets are denoted by capital letters $A, B, C, \ldots$ and their elements are enclosed in curly braces.

Membership notation:

- $x \in A$: x *is* an element of A.
- $x \notin A$: x is *not* an element of A.

Three ways to describe a set:

(i) **Roster (tabular) form**: list all elements, e.g. $A = \{2, 4, 6, 8\}$.

(ii) **Set-builder (rule) form**: $A = \{x : x \text{ is a positive even integer less than } 10\}$.

(iii) **Venn diagram**: pictorial representation (Section 4).

Example 2.1: Convert between forms

(a) $B = \{x : x \text{ is a multiple of } 3,\ 1 \le x \le 15\}$
$\Rightarrow B = \{3, 6, 9, 12, 15\}$.

(b) $C = \{1, 4, 9, 16, 25\}$
$\Rightarrow C = \{x : x = n^2,\ n = 1, 2, 3, 4, 5\}$.

(c) $D = \{a, e, i, o, u\}$
$\Rightarrow D = \{x : x \text{ is a vowel}\}$.

(d) $E = \{x \in \mathbb{Z} : -2 \le x \le 2\}$
$\Rightarrow E = \{-2, -1, 0, 1, 2\}$.

Types of Sets

Type	Description and Example
Empty set $\emptyset$ or $\{\}$	Contains no elements; $n(\emptyset) = 0$.
Universal set S (or U)	The set of all elements under consideration.
Singleton set	Exactly one element, e.g. $\{7\}$.
Finite set	A countable, limited number of elements.
Infinite set	Infinitely many elements, e.g. $\mathbb{N} = \{1, 2, 3, \ldots\}$.
Subset $B \subseteq A$	Every element of B also belongs to A.
Proper subset $B \subsetneq A$	$B \subseteq A$ **and** $B \neq A$.
Power set $\mathcal{P}(A)$	The set of all subsets of A; $n(\mathcal{P}(A)) = 2^{n(A)}$.
Equal sets $A = B$	$A \subseteq B$ **and** $B \subseteq A$ simultaneously.

Important Note

Example 2.2: Subsets and Power Sets

Let $A = \{1, 2, 3\}$**.**

All subsets of A (there are $2^3 = 8$ of them):

$$\emptyset, \quad \{1\}, \quad \{2\}, \quad \{3\}, \quad \{1,2\}, \quad \{1,3\}, \quad \{2,3\}, \quad \{1,2,3\}.$$

So $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$ and $n(\mathcal{P}(A)) = 2^3 = 8$.

Subset checks:

- $\{1,2\} \subseteq A$: **True** (both 1 and 2 are in A).
- $\{1,4\} \subseteq A$: **False** (since $4 \notin A$).
- $A \subseteq A$: **True** (every set is a subset of itself).
- $\emptyset \subseteq A$: **True** (the empty set is a subset of *every* set).
- $\{1,2\} \subsetneq A$: **True** (it is a proper subset because $\{1,2\} \neq A$).

Counting Proper Subsets

A has $2^3 - 1 = 7$ proper subsets (all subsets except A itself).

1. Let $S = \{1, 2, \ldots, 10\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8, 10\}$, $C = \{3, 6, 9\}$.
   (a) Is $C \subseteq A$? Justify. (b) Is $A \subseteq S$?
   (c) List all elements of $\mathcal{P}(C)$ and verify $n(\mathcal{P}(C)) = 2^{n(C)}$.
2. Express each set in set-builder notation:
   (a) $\{2, 4, 8, 16, 32\}$ (b) $\{-3, -2, -1, 0, 1, 2, 3\}$ (c) $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}\right\}$.
3. How many subsets does a set with 6 elements have? How many are proper subsets?
4. If $A = \{p, q, r, s\}$ and $B = \{q, r\}$, determine whether $B \subseteq A$, $B \subsetneq A$, or $B = A$, with justification.
5. A market trader in Kejetia sells three types of fabric: Kente (K), Batik (B), and Adinkra (A). On a given day the trader's stock is: $K = \{\text{red, gold, green}\}$, $B = \{\text{red, blue, white}\}$, $A = \{\text{gold, black}\}$.
   (a) What is the universal set S? (b) List all subsets of A.

The Six Fundamental Set Operations I

Let S denote the universal set and $A, B \subseteq S$.

Complement

$A^c = \{x \in S : x \notin A\}$
Key facts: $(A^c)^c = A$; $S^c = \emptyset$; $\emptyset^c = S$;
$n(A^c) = n(S) - n(A)$.

Union

$A \cup B = \{x : x \in A \text{ or } x \in B\}$
In probability: event that **at least one** of A, B occurs.

Intersection

$A \cap B = \{x : x \in A \text{ and } x \in B\}$
In probability: event that **both** A and B occur simultaneously.

Mutually Exclusive (Disjoint)

A and B are **mutually exclusive** if $A \cap B = \emptyset$.
More generally, $A_1, \ldots, A_n$ are mutually exclusive if $A_i \cap A_j = \emptyset$ for all $i \neq j$.

Set Difference

$A \setminus B = A \cap B^c = \{x : x \in A \text{ and } x \notin B\}$

Symmetric Difference

$A \triangle B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$
Elements in A or B but *not both*.

Algebraic Properties of Set Operations I

Property	**Union form**	**Intersection form**
Commutativity	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributivity	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$A \cup \emptyset = A$	$A \cap S = A$
Domination	$A \cup S = S$	$A \cap \emptyset = \emptyset$
Idempotency	$A \cup A = A$	$A \cap A = A$
Complement	$A \cup A^c = S$	$A \cap A^c = \emptyset$
Double complement	$(A^c)^c = A$	—

Algebraic Properties of Set Operations II

De Morgan's Laws (especially important in probability)

$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

"Neither A nor B occurs" $= (A \cup B)^c = A^c \cap B^c$.
"Not both A and B" $= (A \cap B)^c = A^c \cup B^c$.

Example 3.1: Full Set Operations Workout

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{5, 6, 7, 8, 9\}$.

(a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$
(b) $A \cap B = \{3, 4, 5\}$
(c) $A^c = \{6, 7, 8, 9, 10\}$
(d) $B^c = \{1, 2, 8, 9, 10\}$
(e) $A \setminus B = \{1, 2\}$
(f) $B \setminus A = \{6, 7\}$

(a) $A \triangle B = \{1, 2, 6, 7\}$
(b) $(A \cup B)^c = \{8, 9, 10\}$
(c) *Verify De Morgan:*
$A^c \cap B^c = \{6, 7, 8, 9, 10\} \cap \{1, 2, 8, 9, 10\}$
$= \{8, 9, 10\}$ ✓
(d) $A \cap (B \cup C) = A \cap \{3, 4, 5, 6, 7, 8, 9\} = \{3, 4, 5\}$
(e) $(A \cap B) \cup (A \cap C) = \{3, 4, 5\} \cup \{5\} = \{3, 4, 5\}$
✓ *(distributivity)*

Example 3.2: Integer Intervals

Let $A_1 = \{n \in \mathbb{Z} : 2 \leq n < 15\}$ and $A_2 = \{n \in \mathbb{Z} : 5 < n < 30\}$.

Union and Intersection of Integer Intervals

$$A_1 \cup A_2 = \{n \in \mathbb{Z} : 2 \leq n < 30\}$$

$$A_1 \cap A_2 = \{n \in \mathbb{Z} : 6 \leq n < 15\}$$

Why?

- A_1 contains integers from 2 up to (but not including) 15.
- A_2 contains integers from 6 up to (but not including) 30.
- Their **union** spans from 2 to just below 30.
- Their **intersection** is only where both conditions hold: $n \geq 6$ (from A_2) *and* $n < 15$ (from A_1).

Example 3.3: University Subject Enrolment

A student is selected from a Ghanaian university. Let:

- $M =$ event the student studies Mathematics.
- $T =$ event the student studies Statistics.
- $C =$ event the student studies Computer Science.

Express each event in set notation:

(a) "Studies Mathematics but not Statistics": $M \setminus T = M \cap T^c$.

(b) "Studies at least one of the three subjects": $M \cup T \cup C$.

(c) "Studies all three subjects": $M \cap T \cap C$.

(d) "Studies exactly one subject": $(M \cap T^c \cap C^c) \cup (M^c \cap T \cap C^c) \cup (M^c \cap T^c \cap C)$.

(e) "Studies none of the three subjects": $(M \cup T \cup C)^c = M^c \cap T^c \cap C^c$ (by De Morgan's Law).

Note on (d) and (e)

(d) uses **three mutually exclusive** events. (e) applies **De Morgan's Law** to three sets: $(A \cup B \cup C)^c = A^c \cap B^c \cap C^c$.

Example 3.4: Applying De Morgan's Laws

In the context of a health survey, let:

- H = "patient has hypertension".
- D = "patient has diabetes".

Translating English into Set Notation

"Patient has **neither** condition"	$= (H \cup D)^c = H^c \cap D^c$
"Patient does **not** have both"	$= (H \cap D)^c = H^c \cup D^c$
"Patient has hypertension or is not diabetic"	$= H \cup D^c$

De Morgan's Laws in Probability

$P(\text{neither } H \text{ nor } D) = P(H^c \cap D^c) = P((H \cup D)^c) = 1 - P(H \cup D)$.
This is the **complementary** approach, which often simplifies calculation greatly.

Section 3 Exercises

1. Let $S = \{1, \ldots, 12\}$, $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$, $C = \{1, 2, 3, 4, 5, 6\}$. Find:
   (a) $A \cup B$ (b) $A \cap B$ (c) A^c (d) $B \setminus A$ (e) $A \triangle B$
   (f) $(A \cup B)^c$ (g) $A \cap (B \cup C)$ (h) $(A \cap B) \cup (A \cap C)$. Verify (g)=(h).
2. Let $A = \{5, 1, 0, -3, 6, 8, 9\}$, $B = \{4, 1, 2\}$, $C = \{4, 3, 6, 8\}$. Find:
   (a) $A \cup B$ (b) $B \cup C$ (c) $A \cap (B \cap C)$ (d) $A \triangle C$.
3. Using De Morgan's Laws, simplify:
   (a) $(A \cup B \cup C)^c$ (b) $(A^c \cap B^c)^c$ (c) $A^c \cup B^c$ in terms of $A \cap B$.
4. In a health survey, F="respondent exercises regularly", G="respondent maintains a balanced diet". Describe in words: (a) $F \cup G$ (b) $F \cap G$ (c) $F^c \cap G$ (d) $(F \cup G)^c$ (e) $F \triangle G$.
5. (*Advanced*) Let $A_i = \left\{v : 1 - \frac{1}{2^{i-1}} \le v \le 1 - \frac{1}{2^i}\right\}$, $i = 1, 2, 3, \ldots$
   Determine $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$.
6. A telecom company classifies subscribers. Let V=voice, D=data, M=mobile money. Write set expressions for: (a) voice and data but not mobile money; (b) at least two services; (c) exactly one service; (d) none of the services.

Introduction to Venn Diagrams

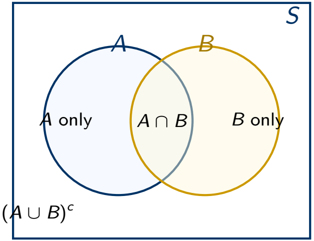
A **Venn diagram** is a pictorial tool for representing sets and their relationships.

Conventions:

- The **universal set** S: drawn as a *rectangle*.
- Each set A, B, C: drawn as a *circle* inside the rectangle.
- Overlapping regions: elements common to those sets.
- Each region is labelled with the *number of elements*.

Two sets A and B partition S into **four disjoint regions**:

1. $A \cap B^c$ (in A only)
2. $A \cap B$ (in both)
3. $A^c \cap B$ (in B only)
4. $A^c \cap B^c$ (in neither)



Inclusion–Exclusion Principle: Two Sets

Inclusion–Exclusion Principle for Two Sets

For any two sets A and B:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

Equivalently: $n(A \cap B) = n(A) + n(B) - n(A \cup B)$.

Why subtract? When we add $n(A)$ and $n(B)$, elements in $A \cap B$ are counted *twice* (once for each set). We subtract $n(A \cap B)$ once to correct for the double-counting.

Example 4.1: Mobile Network Subscribers (Accra)

In a survey of 500 households: 320 subscribe to MTN, 250 to Telecel, 120 to both.

$$n(\text{MTN} \cup \text{Telecel}) = 320 + 250 - 120 = 450.$$

Neither: $500 - 450 = \mathbf{50}$ households.

Examples 4.2 & 4.3: Two-Set Problems

Example 4.2: Investment Club

An investment club has 300 members; 104 invest in stocks, 83 in bonds, and 150 invest in stocks *or* bonds (or both). How many invest in *both*?

$$n(A \cap B) = n(A) + n(B) - n(A \cup B) = 104 + 83 - 150 = \mathbf{37}.$$

Example 4.3: University Elective Courses (60 students)

35 take Economics (E), 30 take Geography (G), 15 take both.

a. $n(E \cup G) = 35 + 30 - 15 = \mathbf{50}$.
b. $n(E \setminus G) = 35 - 15 = \mathbf{20}$ (Economics only).
c. $n(G \setminus E) = 30 - 15 = \mathbf{15}$ (Geography only).
d. Neither: $60 - 50 = \mathbf{10}$.

Check: $20 + 15 + 15 + 10 = 60$. ✓

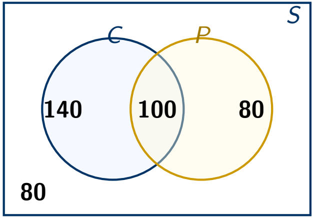
Case Study 2: Healthcare Access in a Ghanaian District

Setup

A community health survey in a rural district covered **400 households**. 240 accessed a health clinic (C), 180 accessed a pharmacy (P), and 80 used *neither*.

Solution:

- Total using at least one: $400 - 80 = 320$.
- $n(C \cap P) = n(C) + n(P) - n(C \cup P) = 240 + 180 - 320 = \mathbf{100}$.
- Clinic only: $240 - 100 = \mathbf{140}$ households.
- Pharmacy only: $180 - 100 = \mathbf{80}$ households.

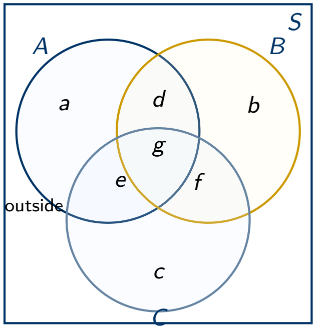


Policy Implication

This analysis helps policymakers understand healthcare access gaps. For example, 140 households rely *only* on the clinic, suggesting the need for more pharmacy services in the area.

Three-Set Venn Diagrams

Three overlapping circles create **seven regions** inside the rectangle, plus one outside.



Region	Meaning
a	in A only
b	in B only
c	in C only
d	in A and B, not C
e	in A and C, not B
f	in B and C, not A
g	in all three: $A \cap B \cap C$
outside	in none of the three

Inclusion–Exclusion: Three Sets

Three-Set Inclusion–Exclusion Principle

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Mnemonic: Add the three singles, subtract the three pairs, add back the triple.

Recovering individual regions from the formula:

- $g = n(A \cap B \cap C)$
- $d = n(A \cap B) - g, \quad e = n(A \cap C) - g, \quad f = n(B \cap C) - g$
- $a = n(A) - d - e - g, \quad b = n(B) - d - f - g, \quad c = n(C) - e - f - g$
- $\text{outside} = n(S) - (a + b + c + d + e + f + g)$

Always verify your work!

$a + b + c + d + e + f + g + \text{outside} = n(S)$

Example 4.4: UG Subject Survey

A survey of **270 UG students**:

Subject	Count	Combination	Count
Mathematics (M)	175	$M \cap T$	135
Statistics (T)	210	$M \cap C$	92
Comp. Sci. (C)	110	$T \cap C$	67
All three $M \cap T \cap C$	62		

$$n(M \cup T \cup C) = 175 + 210 + 110 - 135 - 92 - 67 + 62 = \mathbf{263}.$$

None: $270 - 263 = \mathbf{7}$ students.

Check (fill all 7 regions): $g = 62$; $d = 73$; $e = 30$; $f = 5$; $a = 10$; $b = 70$; $c = 13$.
Total: $10 + 70 + 13 + 73 + 30 + 5 + 62 = 263$. ✓

Example 4.5: Faculty Recreational Activities

A group of **86 faculty families** carpool to baseball (B), opera (O), and theatre (T) each month. Given: 11 attend none; $n(B) = 33$, $n(O) = 35$, $n(T) = 39$; only-B$= 14$, only-O$= 15$, only-T$= 17$.

Find (a) $n(B \cap O \cap T)$**; (b)** $n(B \cap O \cap T^c)$ **(baseball and opera but not theatre).**

Solution. Total attending at least one $= 86 - 11 = 75$. Let $g = n(B \cap O \cap T)$.

$$d_{BO} + d_{BT} = 33 - 14 - g = 19 - g$$

$$d_{BO} + d_{OT} = 35 - 15 - g = 20 - g$$

$$d_{BT} + d_{OT} = 39 - 17 - g = 22 - g$$

Adding all three: $2(d_{BO} + d_{BT} + d_{OT}) = 61 - 3g$.
Also: $14 + 15 + 17 + (d_{BO} + d_{BT} + d_{OT}) + g = 75 \;\Rightarrow\; d_{BO} + d_{BT} + d_{OT} = 29 - g$.
So $2(29 - g) = 61 - 3g \;\Rightarrow\; 58 - 2g = 61 - 3g \;\Rightarrow\; \mathbf{g} = \mathbf{3}$.
Then: $d_{OT} = 10$, $d_{BT} = 9$, $d_{BO} = 7$.

(a) $n(B \cap O \cap T) = \mathbf{3}$. **(b)** Baseball and opera only: $d_{BO} = \mathbf{7}$.

Case Study 3: Banking Channel Usage in Accra

A bank surveyed **300 customers**: Mobile App (M), Internet Banking (I), ATM (A).
$n(M) = 180$, $n(I) = 150$, $n(A) = 120$; $n(M \cap I) = 80$, $n(M \cap A) = 60$, $n(I \cap A) = 50$; $n(M \cap I \cap A) = 30$.

(a) **At least one channel:** $n(M \cup I \cup A) = 180 + 150 + 120 - 80 - 60 - 50 + 30 = \mathbf{290}$.

(b) **None:** $300 - 290 = \mathbf{10}$ customers.

(c) **Exactly one channel:**
Only M $= 180 - 80 - 60 + 30 = 70$; Only I $= 150 - 80 - 50 + 30 = 50$; Only A $= 120 - 60 - 50 + 30 = 40$.
Total $= \mathbf{160}$ customers.

(d) **Exactly two channels:**
$(M \cap I)$ only $= 80 - 30 = 50$; $(M \cap A)$ only $= 60 - 30 = 30$; $(I \cap A)$ only $= 50 - 30 = 20$.
Total $= \mathbf{100}$ customers.

Verification

$160 + 100 + 30 + 10 = 300$. ✓
This kind of analysis guides banks in understanding which digital channels customers use and where to invest resources.

Section 4 Exercises (Part 1)

1. In a survey of 200 first-year UG students: 120 study Economics (E), 80 study Mathematics (M), 95 study English (L); 42 study both E and M, 35 study both M and L, 55 study both E and L, and 20 study all three.
   (a) Draw and label a Venn diagram filling in all seven regions.
   (b) How many study at least one subject?
   (c) How many study exactly one subject?
   (d) How many study none of the three?
2. A school of 85 students: 45 play football (F), 38 basketball (K), 20 cricket (R). Further: $n(F \cap K) = 15$, $n(F \cap R) = 10$, $n(K \cap R) = 8$, $n(F \cap K \cap R) = 5$.
   (a) How many play at least one sport?
   (b) How many play exactly two sports?
   (c) How many play none?
3. In a class of 100 students: 65 passed Mathematics, 45 English, 40 Science; pairwise: 25 (M&E), 20 (M&S), 15 (E&S); 10 passed all three.
   (a) Passed at least one. (b) Passed exactly two. (c) Failed all three.

Section 4 Exercises (Part 2)

- A bank in Accra surveyed 300 customers: 180 use mobile app (M), 150 internet banking (I), 120 ATM (A); $n(M \cap I) = 80$, $n(M \cap A) = 60$, $n(I \cap A) = 50$, $n(M \cap I \cap A) = 30$.
  Verify all counts from Case Study 3, then find: (a) at least one channel; (b) exactly one; (c) none.
- (*Case Study*) A regional health authority surveyed 500 adults in the Ashanti Region: $n(E) = 310$, $n(N) = 260$, $n(A) = 190$; $n(E \cap N) = 160$, $n(E \cap A) = 95$, $n(N \cap A) = 80$; $n(E \cap N \cap A) = 50$.
  (a) How many have at least one healthy behaviour?
  (b) How many have all three?
  (c) How many have none of the three?
  (d) How many have exactly two healthy behaviours?
  (e) Sketch the Venn diagram with region counts filled in.
- (*Data inconsistency challenge*) A researcher claims: 500 Accra households, 320 own a TV (T), 275 a refrigerator (R), 180 an air conditioner (A), 150 own T and R, 90 own T and A, 85 own R and A, 60 own all three.
  Apply inclusion–exclusion. What is $n(T \cup R \cup A)$? Is the data consistent? What does the result imply?

The Multiplication (Fundamental Counting) Principle

The Multiplication Principle

If a task consists of k steps, with step i performable in m_i ways (regardless of earlier choices), then the total number of ways to complete the task is:

$$m_1 \times m_2 \times \cdots \times m_k.$$

Example 5.1: Applications:

- (a) **Ghanaian vehicle plates** (2 letters, then 4 digits, with repetition): $26^2 \times 10^4 = 676 \times 10{,}000 = \mathbf{6{,}760{,}000}$ possible plates.
- (b) **Chop-bar menu** (4 soups, 5 main dishes, 3 drinks): $4 \times 5 \times 3 = \mathbf{60}$ combinations.
- (c) **Coin and die**: $2 \times 6 = \mathbf{12}$ outcomes.
- (d) **Mobile Money PIN** (4 digits, with repetition): $10^4 = \mathbf{10{,}000}$; without repetition: $10 \times 9 \times 8 \times 7 = \mathbf{5{,}040}$.
- (e) **Telecom codes** (3 letters then 4 digits, with repetition): $26^3 \times 10^4 = 17{,}576 \times 10{,}000 = \mathbf{175{,}760{,}000}$.

The Addition Principle

The Addition Principle

If a task can be performed in one of k *mutually exclusive* ways, with the i-th way having m_i options, then the total number of ways to perform the task is:

$$m_1 + m_2 + \cdots + m_k.$$

The key phrase is *mutually exclusive*: **only one alternative is chosen**, not a combination.

Example 5.2: Travel from Accra to Kumasi

Available options: 5 bus companies, 2 airlines, or 1 train service.
Total ways: $5 + 2 + 1 = \mathbf{8}$. (Passenger chooses *one* mode.)

Example 5.3: Exam Question Choice

A student must choose 1 question from Section A (4 questions) *and* 1 from Section B (6 questions).
Total choices: $4 \times 6 = \mathbf{24}$ (multiplication, since *both* choices are made).

Case Study 4: Electoral Candidate Codes

Setup

The Electoral Commission assigns a unique code to each candidate:

- A 1-letter prefix indicating the region (10 regions, different letter each).
- A 3-digit number (000 to 999) for the candidate within the region.

Total possible codes: $10 \times 1{,}000 = \mathbf{10{,}000}$.

Modified rule: If the last digit of the number must be non-zero (to avoid ambiguity with leading zeros), the 3-digit part has $10 \times 10 \times 9 = 900$ options, giving:

$$10 \times 900 = \mathbf{9{,}000} \text{ valid codes.}$$

Addition vs. Multiplication: The Key Distinction

Addition: the task is done by choosing ONE of several alternatives (OR).
Multiplication: the task requires doing step 1 AND step 2 AND ...

Section 5 Exercises

1. A restaurant menu lists 6 starters, 8 main courses, and 4 desserts. How many ways can a diner choose one item from each category?
2. How many different 3-digit numbers can be formed from $\{1, 2, 3, 4, 5\}$:
   (a) with repetition; (b) without repetition?
3. A committee consists of a chairperson, secretary, and treasurer, chosen from 5 men and 4 women with no restrictions. How many ways can the three posts be filled?
4. A password is 4 characters, chosen from 26 letters and 10 digits. How many passwords are possible if: (a) repetition is allowed; (b) no repetition; (c) first character must be a letter (repetition allowed)?
5. A vehicle registration authority issues plates: 2 letters, then a hyphen, then 3 digits, with repetition of letters and digits allowed. How many distinct plates are possible?
6. (*Case Study*) A national examination body prints question paper codes using 2 letters from A–F followed by 2 digits from 1–5. How many distinct codes are possible if: (a) repetition is allowed throughout; (b) no letter may be repeated (digits may repeat); (c) no letter and no digit may be repeated?

What is a Permutation?

A **permutation** is an *ordered* arrangement of objects. Two arrangements are considered *different* if the objects appear in a different order.

The Key Question for Permutations

Does order matter? If YES $\rightarrow$ use a permutation.

Arranging n Distinct Objects (All of Them)

The number of ways to arrange all n distinct objects in a row is:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1, \qquad 0! = 1.$$

Example 6.1:

- (a) Word `SHARP` (5 distinct letters): $5! =$ **120** arrangements.
- (b) Seating 6 students in a row: $6! =$ **720** ways.
- (c) Arranging 7 different books on a shelf: $7! =$ **5,040**.
- (d) 8 athletes in a race (all distinct finishing orders): $8! =$ **40,320**.

Factorial Arithmetic

Example 6.2: Factorial simplifications:

(a) $\dfrac{8!}{6!} = 8 \times 7 = \mathbf{56}$

(b) $\dfrac{10!}{7!\,3!} = \dfrac{10 \times 9 \times 8}{3!} = \dfrac{720}{6} = \mathbf{120}$

(c) $\dfrac{n!}{(n-2)!} = n(n-1)$

(a) $6! - 5! = 720 - 120 = \mathbf{600}$

(b) $\dfrac{(n+1)!}{(n-1)!} = (n+1) \cdot n$

Key Technique: Cancellation

Always cancel before multiplying out:

$$\frac{10!}{7!} = \frac{10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!}} = 10 \times 9 \times 8 = 720.$$

This avoids computing huge intermediate numbers.

Partial Permutations: r Objects from n

Often we arrange only r of the n available objects.

Partial Permutations ${}_nP_r$

The number of ordered arrangements of r objects chosen from n distinct objects (without replacement) is:

$${}_nP_r = \frac{n!}{(n-r)!} = n(n-1)(n-2)\cdots(n-r+1), \quad 0 \le r \le n.$$

Special cases: ${}_nP_0 = 1$ and ${}_nP_n = n!$.

Example 6.3:

a. Elect president, VP, secretary from 10 club members: ${}_{10}P_3 = 10 \times 9 \times 8 = \mathbf{720}$.
b. 5-digit numbers from $\{1, \ldots, 7\}$ without repetition: ${}_7P_5 = 7 \times 6 \times 5 \times 4 \times 3 = \mathbf{2{,}520}$.
c. Gold, silver, bronze medals from 8 athletes: ${}_8P_3 = 8 \times 7 \times 6 = \mathbf{336}$.
d. 4 candidates shortlisted for interview (order matters) from 10: ${}_{10}P_4 = 10 \times 9 \times 8 \times 7 = \mathbf{5{,}040}$.

Permutations with Repeated Objects

Permutations of Non-Distinct Objects

The number of distinct arrangements of n objects, with n_1 of one type, n_2 of a second type, ..., n_k of the k-th type $(n_1 + \cdots + n_k = n)$, is:

$$\frac{n!}{n_1!\, n_2!\, \cdots\, n_k!}.$$

Example 6.4: Word arrangements:

(a) DISTINCT (I×2, T×2): $\frac{8!}{2!\,2!} = \mathbf{10{,}080}$

(b) CARRIER (R×3): $\frac{7!}{3!} = \mathbf{840}$

(c) STATISTICS (S×3, T×3, I×2): $\frac{10!}{3!\,3!\,2!} = \mathbf{50{,}400}$

(a) PROBABILITY (B×2, I×2): $\frac{11!}{2!\,2!} = \mathbf{9{,}979{,}200}$

(b) ACCRA (A×2, C×2): $\frac{5!}{2!\,2!} = \mathbf{30}$

(c) GHANA (A×2): $\frac{5!}{2!} = \mathbf{60}$

Circular Permutations

In a **circular arrangement**, there is no fixed starting point; only the *relative* order of objects matters. One object is fixed to eliminate rotational repetitions.

Circular Permutations

- n distinct objects in a **circle**: $(n-1)!$
- For a **necklace or bracelet** (flip symmetry also counts as same): $\dfrac{(n-1)!}{2}$

Example 6.5:

(a) 5 people at a round table: $(5-1)! = 4! = \mathbf{24}$.
(b) 6 executives at a circular boardroom table: $(6-1)! = 5! = \mathbf{120}$.
(c) A 5-bead bracelet (flip-equivalent): $\dfrac{4!}{2} = \mathbf{12}$.

Why $(n-1)!$ and not $n!$?

In a circle, fixing one person eliminates n rotations that would otherwise give identical arrangements. So we have $n!/n = (n-1)!$ distinct seatings.

Example 6.6: Circular Seating with Restrictions

Five persons A, B, C, D, E sit at a round table.

- (a) **No restriction:** $(5-1)! = 4! = \mathbf{24}$.
- (b) **A and D must sit together:** Treat AD as one unit $\Rightarrow$ 4 units around the table: $(4-1)! \times 2! = 6 \times 2 = \mathbf{12}$.
  (The $\times 2!$ accounts for AD or DA within the block.)
- (c) **C and E must NOT sit together:** Total $-$ (C and E together) $= 24 - 12 = \mathbf{12}$.
  (We use (b)'s logic: if C&E are together, $(4-1)! \times 2! = 12$.)
- (d) **A is fixed at the head of the table:** Fix A's position; remaining 4 people arrange in $4! = \mathbf{24}$ ways.

Key Technique: Complementary Counting

For "not together" restrictions, always compute:
Arrangements where they ARE together, then subtract from total.

Permutations with Restrictions: Summary Table

Restriction	Count
n items, k of one type, rest distinct	$\dfrac{n!}{k!}$
Choose r from n; always include a specific item	$r \cdot {}_{n-1}P_{r-1}$
Choose r from n; one item fixed at a specific position	${}_{n-1}P_{r-1}$
Choose r from n; one item always excluded	${}_{n-1}P_r$
Two specific items always together	Treat as 1 unit, $\times 2!$
Two specific items always apart	Total $-$ (arrangements with them together)
Men and women alternate in a row of $2n$	$2 \times (n!)^2$

Example 6.7: Alternating Gender Seating

4 men and 4 women in a row of 8 chairs.

(a) **No restriction:** $8! = \mathbf{40{,}320}$.

(b) **Men and women alternate** (M W M W ...or W M W M ...):
There are 2 patterns. In each, 4 men fill the men's slots ($4!$ ways) and 4 women fill the women's slots ($4!$ ways):

$$2 \times 4! \times 4! = 2 \times 24 \times 24 = \mathbf{1{,}152}.$$

(c) **All men sit together:** Treat the block of 4 men as one unit among 5 objects (block + 4 women).
Arrange 5 objects: $5!$ ways. Arrange men within the block: $4!$ ways.

$$5! \times 4! = 120 \times 24 = \mathbf{2{,}880}.$$

Example 6.8: Letter Arrangement with Ordering Constraint

The word `YOURSELVES` has 10 letters: Y, O, U, R, S, E, L, V, E, S - with S$\times$2 and E$\times$2.

In how many ways can the letters be arranged if U must immediately precede S ("US" always appears as a block)?

Solution.

- Treat "US" as a single symbol. The 9 items to arrange are: {US, Y, O, R, S, E, L, V, E}.
- Note: one S went into the "US" block, so the remaining S appears once; E still appears *twice*.
- Apply the non-distinct arrangement formula:

$$\frac{9!}{2!} = \frac{362{,}880}{2} = \mathbf{181{,}440}.$$

The 2! accounts for the two E's. The block "US" is already ordered (U before S), so no further division is needed.

Case Study 5: Staff Rostering at a Hospital

Setup

A pharmacy has 8 pharmacists: 3 seniors (S1, S2, S3) and 5 juniors (J1–J5). The department head assigns a *different* pharmacist each day over a 5-day working week (5 of the 8 will be rostered).

- (a) **No restrictions:**
  ${}_8P_5 = 8 \times 7 \times 6 \times 5 \times 4 = \mathbf{6{,}720}$ schedules.
- (b) **S1 must work on Monday:**
  Fix S1 on Monday; arrange 4 from the remaining 7 for the other days:
  ${}_7P_4 = 7 \times 6 \times 5 \times 4 = \mathbf{840}$ schedules.
- (c) **S1 must not work any day this week:**
  Schedule 5 from the remaining 7: ${}_7P_5 = \mathbf{2{,}520}$ schedules.
- (d) **S1 and S2 must work on consecutive days:**
  4 pairs of consecutive days (Mon–Tue, ..., Thu–Fri). For each, S1 and S2 can be ordered in 2! ways, remaining 3 days filled from 6 pharmacists (${}_6P_3 = 120$):

$$4 \times 2 \times 120 = \mathbf{960} \text{ schedules.}$$

Section 6 Exercises (Part 1)

1. Evaluate: (a) $7!$ (b) $\dfrac{9!}{6!}$ (c) $\dfrac{12!}{9!\,3!}$ (d) ${}_8P_3$ (e) ${}_6P_6$.
2. Find the number of arrangements of:
   (a) `KUMASI` (all 6 letters); (b) `ABCDE` taken 3 at a time;
   (c) `MISSISSIPPI` (M$\times$1, I$\times$4, S$\times$4, P$\times$2); (d) `LEGON`.
3. (*MISSISSIPPI check*) Count the repeated letters in MISSISSIPPI. Compute the number of distinct arrangements.
4. In how many ways can 8 students be seated in a row if:
   (a) no restriction; (b) two specific students must sit together;
   (c) two specific students must not sit together?
5. A shelf holds exactly 6 books. In how many ways can 6 books be selected and arranged on it from 10 distinct books?
6. Gold, silver, and bronze medals are awarded to 3 of 12 athletes. In how many ways can the medals be assigned?

Section 6 Exercises (Part 2)

- In how many ways can 6 people be seated at a round table if:
  (a) no restriction; (b) two friends must sit next to each other;
  (c) two rivals must not sit next to each other?
- How many 4-letter code words can be formed from $\{A, B, C, D, E, F\}$ without repetition, if the word must end in a vowel (A or E)?
- Find the number of arrangements of `STATISTICS` in which the three S's are always together. (*Hint: treat SSS as one symbol.*)
- A class of 5 boys and 4 girls lines up for a photograph. How many arrangements if:
  (a) no restriction; (b) all girls stand together; (c) no two girls stand next to each other?
- (*Case Study*) Schedule 6 subjects in 6 periods on Monday:
  (a) no restrictions; (b) Mathematics must be in the first period;
  (c) Mathematics and English must be in adjacent periods;
  (d) Mathematics and Science must not be adjacent.
- (*Case Study*) A radio station has 5 music, 3 news, and 2 advertisement slots. How many row arrangements if:
  (a) no restriction; (b) all news slots together; (c) explain why music/news cannot alternate; find arrangements with them separated; (d) the two advertisement slots at beginning and end.

What is a Combination?

A **combination** is an *unordered* selection of objects. Two selections are the same if they contain exactly the same objects regardless of order.

The Key Question for Combinations

Does order matter? If NO $\rightarrow$ use a combination.

Contrast with permutations. From $\{X, Y, Z\}$, choose 2:

- **Permutations** (order matters): XY, XZ, YX, YZ, ZX, ZY – **6 arrangements**.
- **Combinations** (order irrelevant): $\{X, Y\}, \{X, Z\}, \{Y, Z\}$ – **3 selections**.

Each combination of 2 gives $2! = 2$ permutations, so $\binom{3}{2} = {}_3P_2/2! = 6/2 = 3$.

Combination Formula

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!\,(n-r)!}, \qquad 0 \leq r \leq n.$$

Key Combinatorial Identities

Useful Identities

1. **Symmetry:** $\binom{n}{r} = \binom{n}{n-r}$ (choosing r to include $\equiv$ choosing $n-r$ to exclude)
2. **Pascal's Identity:** $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$
3. **Boundary:** $\binom{n}{0} = \binom{n}{n} = 1$
4. **Single element:** $\binom{n}{1} = n$
5. **Out-of-range:** $\binom{n}{r} = 0$ for $r < 0$ or $r > n$
6. **Relationship to permutations:** ${}_nP_r = r! \cdot \binom{n}{r}$

Example 7.1: Verifying Pascal's Identity ($n = 7,\ r = 3$)

$$\binom{6}{2} + \binom{6}{3} = 15 + 20 = 35 = \binom{7}{3}. \quad \checkmark$$

Worked Examples: Unrestricted Combinations

Example 7.2:

- (a) **Ghana National Lottery (choose 6 from 49):**
  $\binom{49}{6} = 13{,}983{,}816$. $P(\text{win with one ticket}) \approx 7.15 \times 10^{-8}$ – extremely small!
- (b) **Working party of 4 from 7:** $\binom{7}{4} = 35$.
- (c) **Committee of 5 from 15:** $\binom{15}{5} = 3{,}003$.
- (d) **3 balls from 10:** $\binom{10}{3} = 120$.
- (e) **5-card hand from 52-card deck:** $\binom{52}{5} = 2{,}598{,}960$.
- (f) **Mixed selection**: 2 maths books from 8, AND 3 biology books from 6:
  $\binom{8}{2} \times \binom{6}{3} = 28 \times 20 = \mathbf{560}$.
- (g) **Committee of 3 from 5 men and 4 women (no restriction):**
  $\binom{9}{3} = \mathbf{84}$.

Combinations with Restrictions: Summary Table

Restriction	Count
p particular objects always included	$\binom{n-p}{r-p}$
p particular objects always excluded	$\binom{n-p}{r}$
p included *and* q excluded	$\binom{n-p-q}{r-p}$
At least k from a subgroup of size g	$\sum_{i=k}^{\min(r,g)} \binom{g}{i}\binom{n-g}{r-i}$
At most k from a subgroup of size g	$\sum_{i=0}^{k} \binom{g}{i}\binom{n-g}{r-i}$
Two items cannot both be included	Total − (both included)

Example 7.3 & 7.4: Restricted Combinations

Example 7.3: Cricket team (15 players, choose 11):

(a) Player X always chosen: $\binom{14}{10} = \binom{14}{4} = \mathbf{1{,}365}$.

(b) Player X never chosen: $\binom{14}{11} = \binom{14}{3} = \mathbf{364}$.

(c) Captain always in, injured player always out: $\binom{13}{10} = \binom{13}{3} = \mathbf{286}$.

Example 7.4: Parliament committee (10 Majority + 8 Minority, choose 6):

(a) No restriction: $\binom{18}{6} = \mathbf{18{,}564}$.

(b) Equal representation (3 from each): $\binom{10}{3} \times \binom{8}{3} = 120 \times 56 = \mathbf{6{,}720}$.

(c) At least 4 Majority:

$$\binom{10}{4}\binom{8}{2} + \binom{10}{5}\binom{8}{1} + \binom{10}{6}\binom{8}{0} = 5{,}880 + 2{,}016 + 210 = \mathbf{8{,}106}.$$

(d) A named Majority leader must serve: $\binom{17}{5} = \mathbf{6{,}188}$.

Examples 7.5 & 7.6: More Restricted Combinations

Example 7.5: Incompatible Team Members

Choose 4 students from 8 for a tennis team; **Kofi and Ama cannot both be chosen**.
Complementary counting:

$$\binom{8}{4} - \underbrace{\binom{6}{2}}_{\text{both Kofi \& Ama chosen}} = 70 - 15 = \mathbf{55}.$$

Example 7.6: Workplace Groups (7 women, 4 men; groups of 5)

- (a) No restriction: $\binom{11}{5} = \mathbf{462}$.
- (b) Exactly 3 women, 2 men: $\binom{7}{3} \times \binom{4}{2} = 35 \times 6 = \mathbf{210}$.
- (c) At least one man: $\binom{11}{5} - \binom{7}{5} = 462 - 21 = \mathbf{441}$.
- (d) At most one man: $\binom{7}{5} + \binom{4}{1}\binom{7}{4} = 21 + 4(35) = \mathbf{161}$.

Case Study 6: Admission Committees at a Ghanaian University

Setup

12 members: **7 professors** and **5 administrative staff**. A subcommittee of **4** needed.

- (a) **No restriction:** $\binom{12}{4} = \mathbf{495}$ subcommittees.
- (b) **At least 2 professors:**

$$\binom{7}{2}\binom{5}{2} + \binom{7}{3}\binom{5}{1} + \binom{7}{4}\binom{5}{0} = 210 + 175 + 35 = \mathbf{420}.$$

- (c) **The Dean (prof.) and the Registrar (admin.) must both serve:**
  Fix those 2; choose remaining 2 from the other 10: $\binom{10}{2} = \mathbf{45}$.
- (d) **The Dean and the Deputy Registrar cannot both serve:**
  $\binom{12}{4} - \binom{10}{2} = 495 - 45 = \mathbf{450}$.
- (e) **Exactly 1 administrative staff member must serve:**
  $\binom{5}{1} \times \binom{7}{3} = 5 \times 35 = \mathbf{175}$.

Case Study 7: GH¢ Lottery and Probability

Setup

In the **Ghana National Weekly Lottery**, a player selects 6 numbers from 1 to 90.

Total possible selections:

$$\binom{90}{6} = \frac{90 \times 89 \times 88 \times 87 \times 86 \times 85}{720} = \frac{452{,}324{,}547{,}600}{720} = \mathbf{622{,}673{,}538}.$$

Probability of winning the jackpot (all 6 correct) with a single ticket:

$$P(\text{jackpot}) = \frac{1}{622{,}673{,}538} \approx 1.6 \times 10^{-9}.$$

This is approximately **1 in 623 million**: extremely small.

Why Combinatorics Matters in the Real World

Understanding combinations is essential for:

- **Risk assessment**: computing accurate probabilities.
- **Gambling advice**: counselling people on the near-impossibility of winning.

Section 7 Exercises (Part 1)

1. Evaluate: (a) $\binom{10}{4}$ (b) $\binom{12}{5}$ (c) $\binom{8}{8}$ (d) $\binom{15}{2}$ (e) $\binom{7}{0}$.
2. Verify Pascal's Identity for $n = 8$, $r = 3$.
3. A bag contains 10 balls numbered 1–10. Three are drawn without replacement. How many different draws are possible?
4. A committee of 5 from 6 men and 4 women:
   (a) no restriction; (b) one named person is included;
   (c) one named woman is excluded; (d) at least 2 women; (e) at most 1 man?
5. From 5 women and 7 men, form a committee of 2 women and 3 men. How many committees if two specific men refuse to serve together?
6. A mobile money agent sets a 4-digit PIN from 0–9:
   (a) digits may repeat; (b) no digit repeats; (c) no digit repeats and PIN is even.
7. An examination committee of 12 must form a subcommittee of 4:
   (a) total possible; (b) Chairperson and Deputy both serve; (c) they cannot both serve.

Section 7 Exercises (Part 2)

8. In a group of 2 camels, 3 goats, and 10 sheep, choose 6 animals:
   (a) no restriction; (b) both camels included;
   (c) both camels excluded; (d) at most 3 sheep.
8. From a standard 52-card deck, how many 5-card hands contain exactly 2 aces?
8. A parliamentary select committee of 8 from 12 male and 9 female MPs:
   (a) no restriction; (b) at least 3 female MPs; (c) more male than female MPs.
8. (*Case Study*) A software company has 15 developers: 9 senior and 6 junior. Form a project team of 5:
   (a) no restriction; (b) exactly 3 seniors; (c) the team lead (senior) must be included; (d) two particular junior developers cannot both be on the team.
8. (*Case Study*) A returning officer must assign 4 polling agents from a pool of 10 to four specific polling stations (one agent per station). How many assignments are possible? [*Note: since agents are assigned to specific stations, does order matter?*]
8. (*Challenge*) A football coach has 18 players; must select 11. Only 2 of the 18 are goalkeepers; the remaining 10 outfield positions can be filled by any of the other 16.
   (a) How many teams of 11 can be selected?
   (b) How many teams include both goalkeepers? (impossible: explain)
   (c) How many if a specific star player must start?

Summary of Key Formulae (Part 1)

Concept	Formula	When to Use
Factorial	$n! = n(n-1)\cdots 1,\ \ 0! = 1$	Base for all
Non-distinct perm.	$\dfrac{n!}{n_1!\,n_2!\cdots n_k!}$	Arrange n with repeats
Partial permutation	${}_nP_r = \dfrac{n!}{(n-r)!}$	Ordered, r from n
Circular permutation	$(n-1)!$	Arrange n in a circle
Necklace / bracelet	$\dfrac{(n-1)!}{2}$	Circle with flip symmetry
Combination	$\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$	Unordered, r from n

Summary of Key Formulae (Part 2)

Concept	Formula	When to Use
p always included	$\binom{n-p}{r-p}$	Mandatory inclusions
p always excluded	$\binom{n-p}{r}$	Mandatory exclusions
p in & q out	$\binom{n-p-q}{r-p}$	Combined restrictions
I–E (2 sets)	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$	Two overlapping sets
I–E (3 sets)	See full formula	Three overlapping sets
Complementary	Total $-$ unwanted	Often easiest ap-

End-of-Chapter Practice (Q1–Q6)

Q1. **(Sample spaces.)** A student is chosen from a class of 40 (25 female, 15 male; 18 Sciences, 22 Arts). Let $F =$"female" and $T =$"Science student". Describe in words: F^c; $F \cap T$; $F \cup T$; $F^c \cap T^c$.

Q2. **(Set operations.)** Let $S = \{1, \ldots, 20\}$, $A = \{\text{multiples of } 3\}$, $B = \{\text{multiples of } 4\}$, $C = \{\text{odd numbers}\}$. (a) List A, B, C. (b) Find $A \cap B$; $A \cup C$; B^c; $A \setminus C$; $A \triangle B$. (c) Verify De Morgan: $(A \cup B)^c = A^c \cap B^c$.

Q3. **(Inclusion–exclusion.)** 150 final-year students: 80 passed M, 75 E, 60 A; $n(M \cap E) = 40$, $n(M \cap A) = 35$, $n(E \cap A) = 30$, $n(M \cap E \cap A) = 20$. (a) Passed at least one. (b) Passed exactly two. (c) Failed all.

Q4. **(Telecom codes.)** 3 letters (A–Z) then 4 digits (0–9). (a) Total with repetition. (b) No repeated letter, no repeated digit. (c) Starts with G, no repetition.

Q5. **(Permutations.)** Arrangements of `LEGON`: (a) no restriction; (b) begins with L; (c) vowels (E, O) never adjacent.

Q6. **(Word with repeats.)** `PROBABILITY`: (a) no restriction; (b) the two B's appear together.

End-of-Chapter Practice (Q7–Q12)

- **Q7. (Circular.)** 8 executives at a round table: (a) no restriction; (b) CEO and Chairperson sit next to each other; (c) CEO and CFO must not sit next to each other.
- **Q8. (Combinations.)** Committee of 6 from 8 lecturers and 5 students: (a) no restriction; (b) exactly 4 lecturers, 2 students; (c) at least 4 lecturers; (d) a named senior lecturer must serve; (e) a named student must not serve.
- **Q9. (Conflict.)** 4 referees from 10; Kofi and Kwame cannot both be chosen. How many valid panels?
- **Q10. (Mixed.)** Select and *rank* 3 sprinters from 9, and (unranked) choose 4 field athletes from 7. In how many ways?
- **Q11. (Cards.)** From 52 cards: (a) 5-card hands; (b) exactly 3 aces; (c) at least one king; (d) all one suit (flush).
- **Q12. (Banking.)** 15 staff: 9 customer service, 4 tellers, 2 managers. Delegation of 4 to a workshop: (a) no restriction; (b) both managers attend; (c) neither manager; (d) at least one manager; (e) exactly 2 customer service and 2 tellers.

End-of-Chapter Practice (Q13–Q16)

⋆**Q13.** **(Health insurance.)** 600 policyholders: 250 claimed outpatient (O), 180 inpatient (I), 140 dental (D); $n(O \cap I) = 90$, $n(O \cap D) = 60$, $n(I \cap D) = 50$, $n(O \cap I \cap D) = 25$. (a) At least one claim? (b) Exactly one type of claim? (c) No claim? (d) Fraction using dental care only?

⋆**Q14.** **(Exam scheduling.)** Assign 8 courses to 5 time slots (multiple courses per slot allowed). Total unrestricted count: 5^8. [*Hint: this is a surjection / Stirling number problem. Determine the total unrestricted count first.*]

⋆**Q15.** **(Multi-category selection.)** Bookshop: 6 Maths, 8 Statistics, 5 CS texts. Select 5 books. (a) No restriction. (b) At least 2 from each of Maths and Statistics. (c) At most 1 CS text. (d) Exactly 2 Maths, 2 Stats, 1 CS.

⋆**Q16.** **(Security codes.)** 6-character code: first 2 letters (A–Z), last 4 digits (0–9). (a) Repetition allowed? (b) No character repeated (letters and digits treated separately)?

End of Lecture 1

Set Theory and Counting Processes

STAT 112: Introduction to Statistics and Probability I

Department of Statistics and Actuarial Science
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"Good counting is the foundation of good probability."